

**SECTION 1. PRODUCT AND COMPANY IDENTIFICATION**

Product name : ULTRASIL 76

Other means of identification : Not applicable

Recommended use : Cleaning product

Product dilution information : 0,0 % - 0,3 %

Company : Ecolab, S. DE R.L DE C.V.  
Industrial Avenue #28  
Industrial Fractionation, Cuamatla, Cuautitlan Izcalli, Mexico State  
Mexico 54730  
(0155) 5864-1800

Quimiproduktos S. de R.L. de C.V.  
Carretera Miguel Alemán, 990  
Colonia La Encarnación - Apodaca, Nuevo León Mexico Zip Code:  
66633  
(0181) 47395900

Emergency telephone : 01 800 837 2998 Please request assistance in Spanish language

Issuing date : 06.10.2017

**SECTION 2. HAZARDS IDENTIFICATION**
**GHS Classification**
**Product AS SOLD**

|                                 |      |
|---------------------------------|------|
| Oxidizing liquids, Category 3   | H272 |
| Skin corrosion, Sub-category 1A | H314 |
| Serious eye damage, Category 1  | H318 |

**Product AT USE DILUTION**

|                                 |      |
|---------------------------------|------|
| Skin corrosion, Sub-category 1A | H314 |
| Serious eye damage, Category 1  | H318 |

**GHS label elements**
**Product AS SOLD**

Hazard pictograms :



Signal Word : Danger

|                   |        |                                          |
|-------------------|--------|------------------------------------------|
| Hazard Statements | : H272 | May intensify fire; oxidizer.            |
|                   | : H314 | Causes severe skin burns and eye damage. |

|                          |                      |                                                                                                |
|--------------------------|----------------------|------------------------------------------------------------------------------------------------|
| Precautionary Statements | : <b>Prevention:</b> |                                                                                                |
|                          | P210                 | Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. |
|                          | P220                 | Keep away from clothing and other combustible materials.                                       |
|                          | P264                 | Wash skin thoroughly after handling.                                                           |
|                          | P280                 | Wear protective gloves/ protective clothing/ eye                                               |

# SAFETY DATA SHEET

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protection/ face protection.

### Response:

P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.

P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water.

P304 + P340 + P310 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Immediately call a POISON CENTER or doctor/ physician.

P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/ physician.

P363 Wash contaminated clothing before reuse.

P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.

### Storage:

P405 Store locked up.

### Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

## Product AT USE DILUTION

Hazard pictograms :



Signal Word : Danger

Hazard Statements : H314 Causes severe skin burns and eye damage.

Precautionary Statements : **Prevention:**

P264 Wash skin thoroughly after handling.

P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.

### Response:

P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.

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P405 Store locked up.

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P501 Dispose of contents/ container to an approved waste disposal plant.

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### Product AS SOLD

**Other hazards** : Do not mix with bleach or other chlorinated products – will cause chlorine gas.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

### Product AS SOLD

Pure substance/mixture : Mixture

| Chemical name | CAS-No.   | Concentration (%) |
|---------------|-----------|-------------------|
| nitric acid   | 7697-37-2 | 30 - 50           |

### Product AT USE DILUTION

No hazardous ingredients

## SECTION 4. FIRST AID MEASURES

### Product AS SOLD

In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.

In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.

If swallowed : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.

Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Notes to physician : Treat symptomatically.

Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

### Product AT USE DILUTION

In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.

In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.

If swallowed : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.

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### SECTION 5. FIRE-FIGHTING MEASURES

#### Product AS SOLD

- Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
- Unsuitable extinguishing media : None known.
- Specific hazards during fire fighting : Oxidizer. Contact with other material may cause fire.
- Hazardous combustion products : Decomposition products may include the following materials:  
Carbon oxides  
Nitrogen oxides (NOx)  
Sulfur oxides  
Oxides of phosphorus
- Special protective equipment for fire-fighters : Use personal protective equipment.
- Specific extinguishing methods : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

### SECTION 6. ACCIDENTAL RELEASE MEASURES

#### Product AS SOLD

- Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.
- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

#### Product AT USE DILUTION

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material to ensure runoff does not reach a waterway.

#### SECTION 7. HANDLING AND STORAGE

##### Product AS SOLD

Advice on safe handling : Do not ingest. Do not get in eyes, on skin, or on clothing. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not mix with bleach or other chlorinated products – will cause chlorine gas.

Conditions for safe storage : Keep in a cool, well-ventilated place. Keep away from reducing agents. Keep away from strong bases. Keep away from combustible material. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

Storage temperature : -30 °C to 50 °C

##### Product AT USE DILUTION

Advice on safe handling : Do not ingest. Do not get in eyes, on skin, or on clothing. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling.

Conditions for safe storage : Keep away from strong bases. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

#### SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

##### Product AS SOLD

##### Ingredients with workplace control parameters

| Ingredients | CAS-No.   | Form of exposure | Permissible concentration | Basis             |
|-------------|-----------|------------------|---------------------------|-------------------|
| Nitric acid | 7697-37-2 | LMPE-PPT         | 2 ppm<br>5 mg/m3          | MX OEL            |
|             |           | LMPE-CT          | 4 ppm<br>10 mg/m3         | MX OEL            |
|             |           | VLE-PPT          | 2 ppm                     | NOM-010-STPS-2014 |
|             |           | VLE-CT           | 4 ppm                     | NOM-010-STPS-2014 |
| nitric acid | 7697-37-2 | TWA              | 2 ppm                     | ACGIH             |
|             |           | STEL             | 4 ppm                     | ACGIH             |
|             |           | STEL             | 4 ppm<br>10 mg/m3         | NIOSH REL         |
|             |           | TWA              | 2 ppm<br>5 mg/m3          | NIOSH REL         |
|             |           | TWA              | 2 ppm<br>5 mg/m3          | OSHA Z1           |

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

##### Personal protective equipment

Eye protection : Safety goggles  
Face-shield

Hand protection : Wear the following personal protective equipment:

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Standard glove type.

Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

#### Product AT USE DILUTION

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

#### Personal protective equipment

Eye protection : Safety goggles  
Face-shield

Hand protection : Wear the following personal protective equipment:  
Standard glove type.  
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                         | Product AS SOLD                                | Product AT USE DILUTION |
|-----------------------------------------|------------------------------------------------|-------------------------|
| Appearance                              | : liquid                                       | liquid                  |
| Color                                   | : colorless                                    | colorless               |
| Odor                                    | : pungent                                      | odorless                |
| pH                                      | : 1,0, (100 %)                                 | 1,0 - 2,0               |
| Flash point                             | : Not applicable, Does not sustain combustion. |                         |
| Odor Threshold                          | : No data available                            |                         |
| Melting point/freezing point            | : No data available                            |                         |
| Initial boiling point and boiling range | : > 100 °C                                     |                         |
| Evaporation rate                        | : No data available                            |                         |
| Flammability (solid, gas)               | : No data available                            |                         |
| Upper explosion limit                   | : No data available                            |                         |
| Lower explosion limit                   | : No data available                            |                         |

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|                                        |                     |
|----------------------------------------|---------------------|
| Vapor pressure                         | : No data available |
| Relative vapor density                 | : No data available |
| Relative density                       | : 1,21              |
| Water solubility                       | : soluble           |
| Solubility in other solvents           | : No data available |
| Partition coefficient: n-octanol/water | : No data available |
| Autoignition temperature               | : No data available |
| Thermal decomposition                  | : No data available |
| Viscosity, kinematic                   | : No data available |
| Explosive properties                   | : No data available |
| Oxidizing properties                   | : No data available |
| Molecular weight                       | : No data available |
| VOC                                    | : No data available |

### SECTION 10. STABILITY AND REACTIVITY

#### Product AS SOLD

|                                    |                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Chemical stability                 | : Stable under normal conditions.                                                                                                                |
| Possibility of hazardous reactions | : Do not mix with bleach or other chlorinated products – will cause chlorine gas.                                                                |
| Conditions to avoid                | : None known.                                                                                                                                    |
| Incompatible materials             | : Metals<br>Bases<br>Organic materials                                                                                                           |
| Hazardous decomposition products   | : Decomposition products may include the following materials:<br>Carbon oxides<br>Nitrogen oxides (NOx)<br>Sulfur oxides<br>Oxides of phosphorus |

### SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

#### Potential Health Effects

#### Product AS SOLD

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Eyes             | : Causes serious eye damage.                                  |
| Skin             | : Causes severe skin burns.                                   |
| Ingestion        | : Causes digestive tract burns.                               |
| Inhalation       | : May cause nose, throat, and lung irritation.                |
| Chronic Exposure | : Health injuries are not known or expected under normal use. |

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#### Product AT USE DILUTION

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Eyes             | : Causes serious eye damage.                                  |
| Skin             | : Causes severe skin burns.                                   |
| Ingestion        | : Causes digestive tract burns.                               |
| Inhalation       | : May cause nose, throat, and lung irritation.                |
| Chronic Exposure | : Health injuries are not known or expected under normal use. |

#### Experience with human exposure

#### Product AS SOLD

|              |                                 |
|--------------|---------------------------------|
| Eye contact  | : Redness, Pain, Corrosion      |
| Skin contact | : Redness, Pain, Corrosion      |
| Ingestion    | : Corrosion, Abdominal pain     |
| Inhalation   | : Respiratory irritation, Cough |

#### Product AT USE DILUTION

|              |                                 |
|--------------|---------------------------------|
| Eye contact  | : Redness, Pain, Corrosion      |
| Skin contact | : Redness, Pain, Corrosion      |
| Ingestion    | : Corrosion, Abdominal pain     |
| Inhalation   | : Respiratory irritation, Cough |

#### Toxicity

#### Product AS SOLD

#### Product

|                                   |                     |
|-----------------------------------|---------------------|
| Acute oral toxicity               | : No data available |
| Acute inhalation toxicity         | : No data available |
| Acute dermal toxicity             | : No data available |
| Skin corrosion/irritation         | : No data available |
| Serious eye damage/eye irritation | : No data available |
| Respiratory or skin sensitization | : No data available |
| Carcinogenicity                   | : No data available |
| Reproductive effects              | : No data available |
| Germ cell mutagenicity            | : No data available |
| Teratogenicity                    | : No data available |
| STOT-single exposure              | : No data available |
| STOT-repeated exposure            | : No data available |
| Aspiration toxicity               | : No data available |

### SECTION 12. ECOLOGICAL INFORMATION



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#### Product AS SOLD Ecotoxicity

Environmental Effects : This product has no known ecotoxicological effects.

#### Product

Toxicity to fish : 96 h LC50: > 120 mg/l

Toxicity to daphnia and other aquatic invertebrates : 48 h EC50: > 120 mg/l

Toxicity to algae : No data available

#### Persistence and degradability

##### Product AS SOLD

Not applicable - inorganic

##### Product AT USE DILUTION

Not applicable - inorganic

#### Bioaccumulative potential

No data available

#### Mobility in soil

No data available

#### Other adverse effects

No data available

### SECTION 13. DISPOSAL CONSIDERATIONS

#### Product AS SOLD

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

#### Product AT USE DILUTION

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

### SECTION 14. TRANSPORT INFORMATION

#### Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

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**Land transport: generally applicable for transport in Mexico.**

UN number : 2031  
Description of the goods : NITRIC ACID  
Class : 8  
Packing group : II  
Environmentally hazardous : no

#### Air transport (IATA)

Contact Regulatory for air freight eligibility

#### Sea transport (IMDG/IMO)

UN number : 2031  
Description of the goods : NITRIC ACID  
Class : 8  
Packing group : II  
Marine pollutant : no

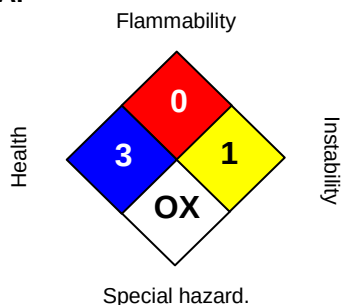
### SECTION 15. REGULATORY INFORMATION

Mexico: Our MSDS meets the requirements established by the Official Mexican Regulation NOM-018-STPS-2015, Harmonized System for the identification and communication of hazards and risks by dangerous chemical substances in work centers.

### SECTION 16. OTHER INFORMATION

#### Product AS SOLD

##### NFPA:



##### HMIS III:

|                 |   |
|-----------------|---|
| HEALTH          | 3 |
| FLAMMABILITY    | 0 |
| PHYSICAL HAZARD | 1 |

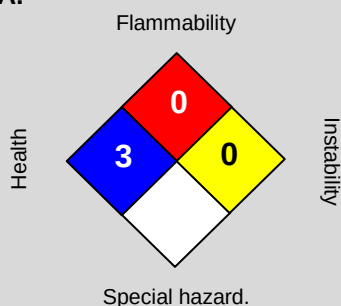
0 = not significant, 1 = Slight,  
2 = Moderate, 3 = High  
4 = Extreme, \* = Chronic

#### Product AT USE DILUTION

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#### NFPA:



#### HMIS III:

|                        |          |
|------------------------|----------|
| <b>HEALTH</b>          | <b>3</b> |
| <b>FLAMMABILITY</b>    | <b>0</b> |
| <b>PHYSICAL HAZARD</b> | <b>0</b> |

0 = not significant, 1 = Slight,  
2 = Moderate, 3 = High  
4 = Extreme, \* = Chronic

Issuing date : 06.10.2017  
Version : 1.1  
Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the MSDS.

The information provided in this Material Safety Data Sheet is considered correct but is not thorough and is intended for guidance only, which is based on current knowledge of the substance or mixture and is applicable to the appropriate safety precautions for the product.